

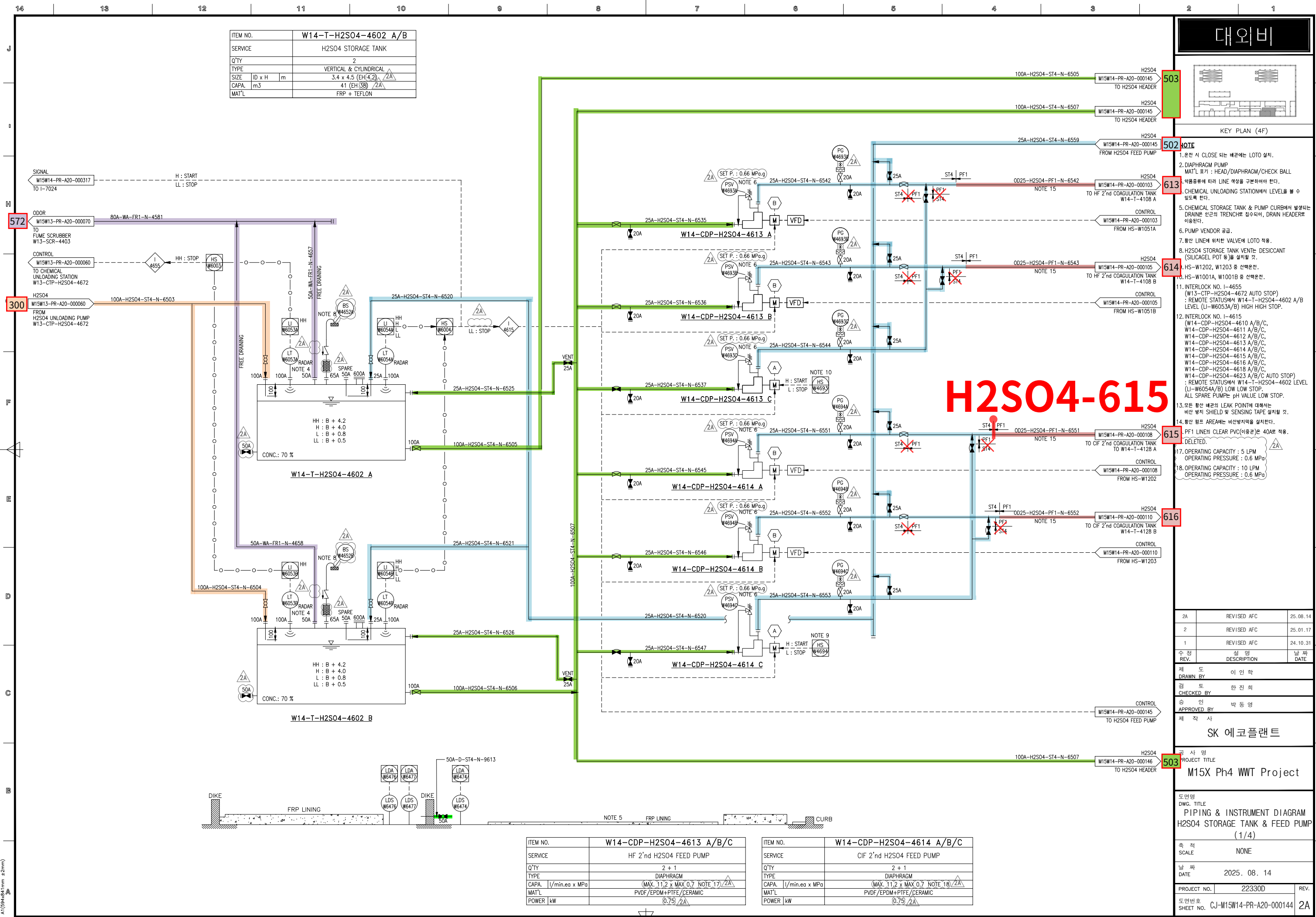
DRAWING LIST FOR TEST PACKAGE

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

H2SO4-615

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	615	4FA	OD25-H2SO4-PF1-N-6551-01	1 OF 2	0	CJ-M15W14-PR-A20-000108/144	1A/2A	
2	615	4FA	OD25-H2SO4-PF1-N-6551-02	2 OF 2	0	CJ-M15W14-PR-A20-000108/144	1A/2A	
3	615	5FA	OD25-H2SO4-PF1-N-6551-01	1 OF 1	0	CJ-M15W14-PR-A20-000108/144	1A/2A	
4	615	RFA	OD25-H2SO4-PF1-E-6551-01	1 OF 2	1	CJ-M15W14-PR-A20-000108/144	1A/2A	
5	615	RFA	OD25-H2SO4-PF1-E-6551-02	2 OF 2	1	CJ-M15W14-PR-A20-000108/144	1A/2A	



ITEM NO.	W14-T-H2SO4-4602 A/B
SERVICE	H2SO4 STORAGE TANK
Q'TY	2
TYPE	VERTICAL & CYLINDRICAL
SIZE	3.4 x 4.5 (EH(4.2) 2A
CAPA.	41 (EH(38) 2A
MAT'L	FRP + TEFLON

대외비

KEY PLAN (4F)

- NOTE
1. 온전 시 CLOSE 되는 배관에는 LOTO 설치.
 2. DIAPHRAGM PUMP MAT'L 표기 : HEAD/DIAPHRAGM/CHECK BALL
 3. 역류종류에 따라 LINE 역상을 구분하여야 한다.
 4. CHEMICAL UNLOADING STATION에서 LEVEL을 볼 수 있도록 한다.
 5. CHEMICAL STORAGE TANK & PUMP CURB에서 발생하는 DRAINE 인근의 TRENCH로 침수되어, DRAIN HEADER로 이용된다.
 6. PUMP VENDOR 공급.
 7. 환산 LINE에 위치한 VALVE에 LOTO 적용.
 8. H2SO4 STORAGE TANK VENT는 DESICCANT (SILICAGEL POT 등)을 설치할 것.
 9. HS-W1202, W1203 중 선택연선.
 10. HS-W1001A, W1001B 중 선택연선.
 11. INTERLOCK NO. I-4655 (W13-CTP-H2SO4-4672 AUTO STOP) : REMOTE STATUS에서 W14-T-H2SO4-4602 A/B LEVEL (LI-W6053A/B) HIGH HIGH STOP.
 12. INTERLOCK NO. I-4615 (W14-CDP-H2SO4-4610 A/B/C, W14-CDP-H2SO4-4611 A/B/C, W14-CDP-H2SO4-4612 A/B/C, W14-CDP-H2SO4-4613 A/B/C, W14-CDP-H2SO4-4614 A/B/C, W14-CDP-H2SO4-4615 A/B/C, W14-CDP-H2SO4-4616 A/B/C, W14-CDP-H2SO4-4618 A/B/C, W14-CDP-H2SO4-4623 A/B/C AUTO STOP) : REMOTE STATUS에서 W14-T-H2SO4-4602 LEVEL (LI-W6054A/B) LOW LOW STOP. ALL SPARE PUMPS는 pH VALUE LOW STOP.
 13. 모든 환산 배관의 LEAK POINT에 대해서는 비산 방지 SHIELD 및 SENSING TAPE 설치할 것.
 14. 환산 펌프 AREA에는 비산방지막을 설치한다.
 15. PF1 LINE의 CLEAR PVC(이용관)은 40A로 적용. DELETED.
 17. OPERATING CAPACITY : 5 LPM OPERATING PRESSURE : 0.6 MPa
 18. OPERATING CAPACITY : 10 LPM OPERATING PRESSURE : 0.6 MPa

2A	REVISED AFC	25.08.14
2	REVISED AFC	25.01.17
1	REVISED AFC	24.10.31
수정	실명	날짜
REV.	DESCRIPTION	DATE

제 도	이 인 학
DRAWN BY	
검 토	한 진 희
CHECKED BY	
승 인	박 동 명
APPROVED BY	

제 작 사
SK 에코플랜트

공 사 명	M15X Ph4 WWT Project
PROJECT TITLE	

도면명	PIPING & INSTRUMENT DIAGRAM
DWG. TITLE	H2SO4 STORAGE TANK & FEED PUMP
속 척	(1/4)
SCALE	NONE
날 짜	2025. 08. 14
DATE	

PROJECT NO.	22330D	REV.	A
도면번호	CJ-M15W14-PR-A20-000144		
SHEET NO.	2A		

ITEM NO.	W14-CDP-H2SO4-4613 A/B/C
SERVICE	HF 2'nd H2SO4 FEED PUMP
Q'TY	2 + 1
TYPE	DIAPHRAGM
CAPA.	1/min.ea x MPa (MAX. 11.2 x MAX 0.7 NOTE 17) 2A
MAT'L	PVDF/EPDM+PTFE/CERAMIC
POWER kW	(0.75) 2A

ITEM NO.	W14-CDP-H2SO4-4614 A/B/C
SERVICE	CIF 2'nd H2SO4 FEED PUMP
Q'TY	2 + 1
TYPE	DIAPHRAGM
CAPA.	1/min.ea x MPa (MAX. 11.2 x MAX 0.7 NOTE 18) 2A
MAT'L	PVDF/EPDM+PTFE/CERAMIC
POWER kW	(0.75) 2A



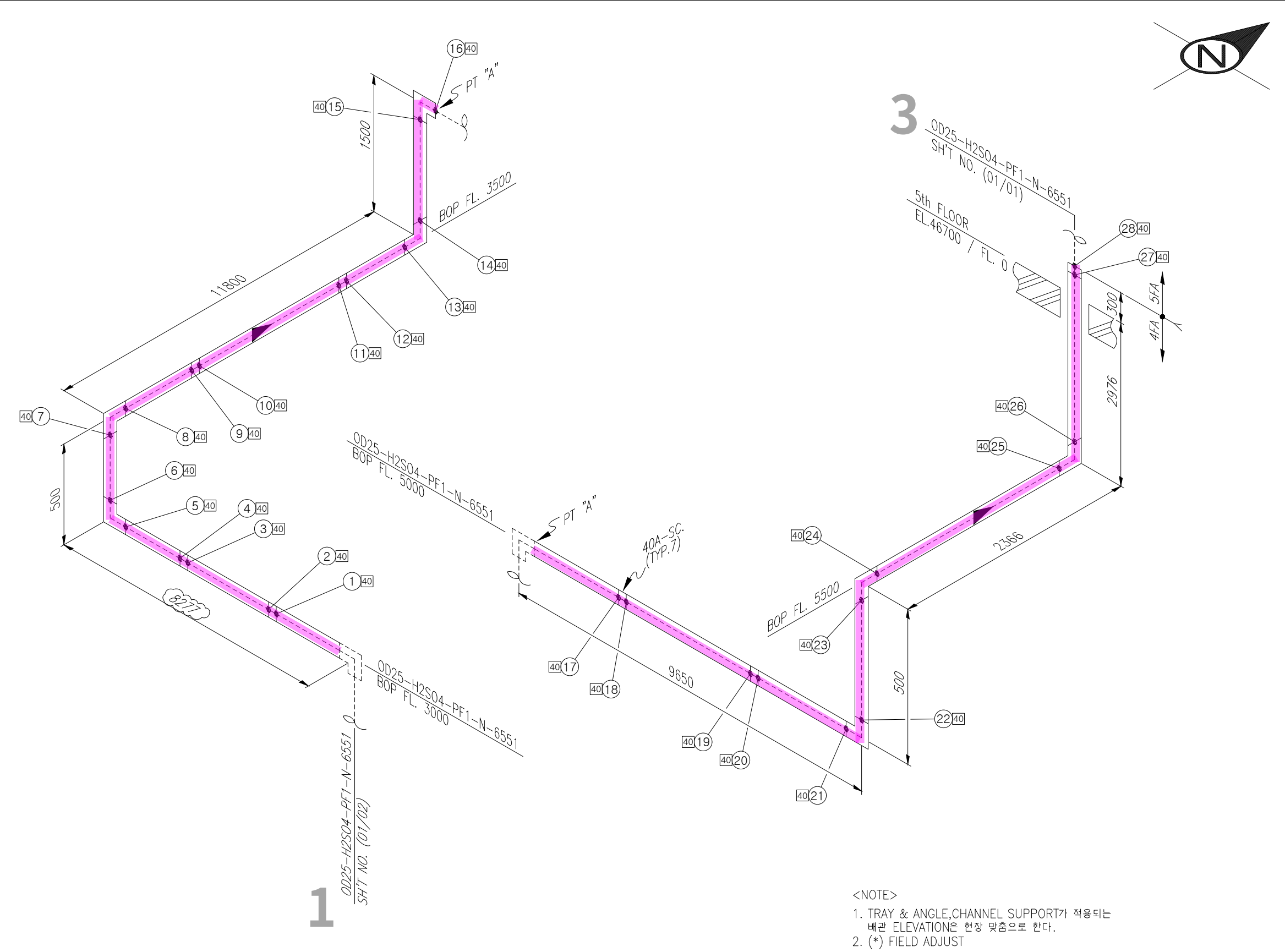
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB10/VB2/VB210	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD10	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			



1

SK ecoplant



* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14	△	25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	CONTRACTOR SK ecoplant	PROJECT NAME M15X Ph4 Project
	THICK.	-	DESIGN	6.9	60	UNIT	4FA								
			OPERATING	6.2	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040008-A									
						REF. P & ID CJ-M15W14-PR-A20-000144		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	COMMODITY H2SO4	JOB NO. 22330D
														LINE NO. OD25-H2SO4-PF1-N-6551	SH'T NO. 02 / 02
															REV. 0

[illegible]

<NOTE>

1. (*) FIELD ADJUST
2. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는
배관 ELEVATION은 현장 맞춤으로 한다.

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* SUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* SUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB2	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD10	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL-PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

CLIENT



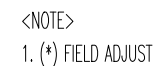
3

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	6.7	60	UNIT 5FA								M15X Ph4 Project		
			OPERATING	6.1	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-050008-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY	H2SO4	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000144	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	0D25-H2SO4-PF1-N-6551	SH'T NO. 01 / 01	REV. 0

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.

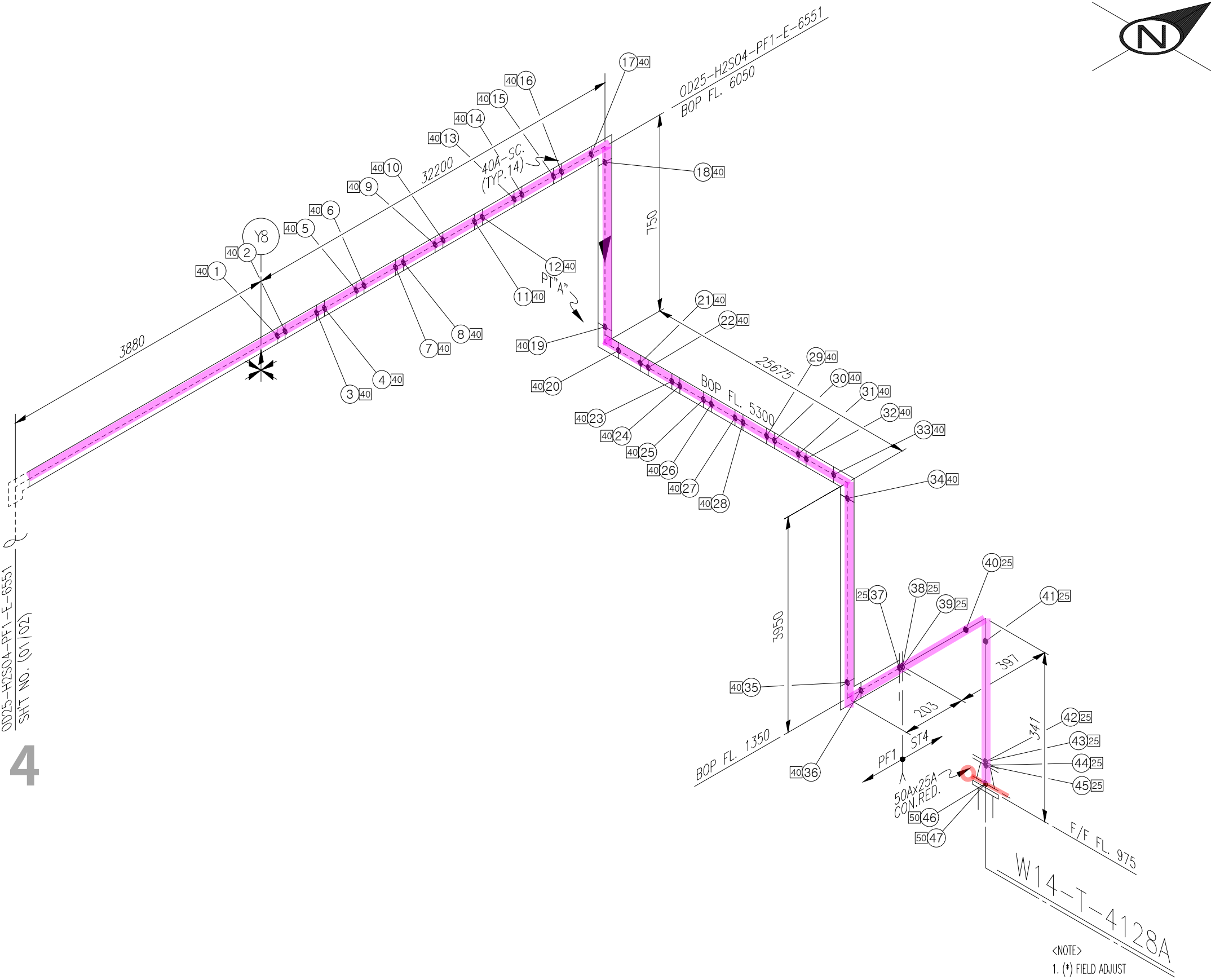


TOR  SK ecoplant



INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	25	DESIGN	6.9	60	UNIT RFA						
			OPERATING	6.2	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-R10008-A		25.03.31	REVISED AS MARKED	황래중	안남후	박동영
						REF. P & ID CJ-M15W14-PR-A20-000108		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영
						REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'.	

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출될 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생한 경우 유포자에게 민/형사상 책임을 요구할 수 있음



4

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14									CONTRACTOR	PROJECT NAME	
	THICK.	25	DESIGN	6.9	60	UNIT	RFA										M15X Ph4 Project	
			OPERATING	6.2	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-R10008-A			25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY		H2SO4	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000108			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.		0D25-H2SO4-PF1-E-6551	SH'T NO. 02 / 02	REV. 1
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D					

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.